Agency loops for students in an age of Agentic AI

Frederik J van Deventer

Introduction

"But Chat said this was a good solution" is something I hear daily in a number of variations from my students, where Chat means ChatGPT or another chat-based LLM project. Students are learning to work with these systems more and more (Kumar et al. 2024) as AI is playing a bigger part in student's academic work, school projects and school policy.

The freedom to act according to one's own intention and volition is often described as agency, or a level of control (Skinner, n.d.). It is this freedom that is described as *achieved agency* by Biesta and Tedder (2007) when it is **realised** "through the active engagement of individuals with aspects of their contexts-for-action", that always exists in an temporal-relational *ecology* (Biesta and Tedder 2006). This *achieved agency* is under pressure when systems circulate responsibility among human and non-human actors, creating "agency loops" (Leonardi 2025). Sartre (1946) said that our freedom "depends entirely upon the freedom of others and that the freedom of others depends upon our own".

Over the last decade the question whether Artificial Intelligence (AI) itself exhibits agency has been debated (Legaspi, He, and Toyoizumi 2019; Swanepoel 2021). This distinction between human and technological agency becomes more blurred with the emerging paradigm of Agentic AI (AAI), which is different to previous forms of AI such as Generative AI (GenAI), which we encounter in platforms like ChatGPT, Claude or Co-pilot (Naik, Naik, and Naik 2024), because AAI can operate (semi-)autonomously and pursue complex goals with little human intervention (Acharya, Kuppan, and Divya 2025). Both GenAI and AAI largely depend on recent developments in Large Language Models (LLMs). With AAI multiple instances of these GenAI-bots run in the background interfacing with eachother creating seemingly autonomous and agentic systems, as a evolutionary next step from the GenAI chat-platforms widely in use among students already (Silvennoinen, Aksovaara, and Alanko-Turunen 2025; Dai 2025).

The *iterational* aspect of Emirbayer and Mische (1998)'s agency upon which Biesta and Tedder (2007) based the *achieved agency* is exposed in structure and collective beliefs, as *truth* expressed in instruments such as policies or punitive action (Foucault 1980, 131), or as expressed by teachers in their narratives or dispositions (Nespor 1987). Beliefs about "inevitability of technology" and the cult of innovation (Winner 2018) pervade, while at the same time mistrust and weariness tries to keeps AI out of universities (Guest et al. 2025). Technology is never just neutral nor is it inherently good or bad (Morrow 2014; Heyndels 2023) it is there to be assistive to our human needs (Arora 2024, 32), and changes us just as much as we change it (Latour 1994, 33). Regulation and policy fueled by ideology (both benefiting the few or the many) has been the driver for the direction this advancement would take us (Johnson and Acemoglu 2023, 57).

Students navigate these punitive faculties, policies and teacher narratives as ecology in which they need to realize goals, their agency changing from situation to situation. Is Agentic AI infringing on their intentional volitions realized as *achieved agency* as the temporal-relational *ecology* changes?

Knowledge Gap

Most literature focusing on AI is not (yet) mentioning AAI, but using GenAI, which is closely related as precursor to AAI paradigm that also utilizes LLMs. This firstly shows the need to further examine the phenomenon of AAI in relation to students. Secondly where GenAI is used in studies we can use as corollary for the type of results AAI.

The way AAI or GenAI is influencing agency in students seems not to have been studied, apart from the influence it has on critical thinking (Suriano et al. 2025), and why and how they use it for studying (Dai 2025; Silvennoinen, Aksovaara, and Alanko-Turunen 2025; Suonpää, Heikkilä, and Dimkar 2024). The Human AI - Interaction framework suggested by Sundar (2020) incorporates agency, but mostly highlights interesting avenues to pursue rather than diving into topics such as: human and AI collaboration, the seeming either/or thinking of machine vs human agency; agency is seen as a "mediating variable" in the machine/human interaction (Sundar 2020).

Agentic AI and AI goals in general, largely seem to focus on increasing abilities for models or replacing tedious tasks performed by humans. While research has been done about future projections of human agency in conjunction with AI in decision making (Pew Research Center, Anderson, and Rainie 2023) and in education (Mouta, Pinto-Llorente, and Torrecilla-Sánchez 2025) it does not deal with how achieved agency for humans now. Or how AAI is changing realized intentions.

Theorizations of agency by Biesta and Tedder (2007) clarify that action flows from the values and beliefs of a person. This study will try to deepen understanding on this interplay of beliefs and agency realised in the context of Agentic AI among students. As action and belief tied together closely (McCormick 2014) it is relevant to investigate these beliefs that inform action further.

Research Questions

Main research question

How is Agentic AI influencing the ecology in which agency-as-achieved can be enacted?

Sub Questions

1. What values and beliefs are prevalent among students concerning Agentic AI?
2. How are prevalent beliefs disclosed within the ecology of achieved agency changing policy and teacher narratives because of AAI?
3. In what ways are students performing day-to-day tasks, by either conforming or subverting rules and restrictions?

Methodology

Table 1: Research Questions and how they relate to methods

	Question	Method	minimum number of participants
RQ1	What values and beliefs are prevalent among students concerning Agentic AI?	Thematic Analysis / UMT	n = 6
RQ2	How are prevalent beliefs disclosed within the ecology of achieved agency changing policy and teacher narratives because of AAI?	Critical Discourse Analyis	n = 5-8
RQ3	In what ways are students performing day-to-day tasks, by either conforming or subverting rules and restrictions?	Mixed-Method	n = 5-8 for interviews n ~ 200 for surveys

RQ1: Systematic Literature Review

As clarified in the Knowledge Gap, there is a close relation between action and belief, however in the context of Agentic AI the values and beliefs among students. In Kramer and Engeström (2019) the word *worldview* is used to describe this belief, which in Biesta and Tedder (2006)'s definition would contribute to the *ecology*. The way this is investigated in Kramer and Engeström (2019) is through the use of Leontiev (2007)'s Ultimate Meanings Technique (UMT).

Where Kramer and Engeström (2019) used UMT to discover worldview among teachers I will investigate among a small group of students.

Thematic analysis will be used as Naeem et al. (2023) and Braun and Clarke (2021) describe it to find common themes and topics in personal qualitative interview with students concerning these worldviews.

After the initial analysis and recognized larger themes questions will be formulated to be able to use a more quantitative approach in the form of a survey.

RQ2: Critical Discourse Analyis

Through *critical discourse analyis* (CDA) I will look at what kind of language is constructing narratives, describes relations of power and how it creates or reinforces beliefs (Blommaert and Bulcaen 2000). With CDA I can take a look at school policy and try to find patterns in power dynamics and social effects in discourse. In further analysis I will analyze context and intertextuality to challenge social inequalities. I will focus mainly on step two: *Identify obstacles to addressing the social wrong* of the CDA process as described by Fairclough (2013).

1. Analyse dialectical relations
2. Select texts
3. Analyse texts interdiscursively and linguistically

To assist in the Critical Discourse analysis I will use the Cui et al. (2023) package for R to help quantify some aspects of it.

Besides school policy, I will sit in in classes to discover what beliefs are disclosed in teacher narratives concerning Agentic AI or GenAI.

RQ3: Mixed-method

To find out where and how this impacts students lives I want to follow the day-to-day tasks of a small number of students to find out where Agentic AI is confronting students with its actions. I will interview a small group of students to uncover what comonalities among students can be found.

Acharya, Deepak Bhaskar, Karthigeyan Kuppan, and B. Divya. 2025. "Agentic AI: Autonomous Intelligence for Complex Goals—a Comprehensive Survey." *IEEE Access : Practical Innovations, Open Solutions* 13: 18912–36. https://doi.org/10.1109/ACCESS.2025.3532853.

Arora, Payal. 2024. *From Pessimism to Promise: Lessons from the Global South on Designing Inclusive Tech*. MIT Press.

Biesta, Gert, and M. Tedder. 2006. "How Is Agency Possible? Towards an Ecological Understanding of Agency-as-Achievement." Working paper 5. The Learning Lives project.

Biesta, Gert, and Michael Tedder. 2007. "Agency and Learning in the Lifecourse: Towards an Ecological Perspective." *Studies in the Education of Adults* 39 (2): 132–49. https://doi.org/10.1080/02660830.2007.11661545.

Blommaert, Jan, and Chris Bulcaen. 2000. "Critical Discourse Analysis." *Annual Review of Anthropology* 29: 447–66. https://www.jstor.org/stable/223428.

Braun, Virginia, and Victoria Clarke. 2021. "Thematic Analysis: A Practical Guide to Understanding and Doing." *Thousand Oaks*.

Cui, Qi, Joshua P. Le, Albert Chai, Andrew S. Lee Without Orcid, Jitarth Sheth Without Orcid, Kevin Banh Without Orcid, Priya Pahal Without Orcid, Katherine Ly Without Orcid, and Stanley M. Lo. 2023. "discourseGT: An R Package to Analyze Discourse Networks in Educational Contexts." *Journal of Open Source Software* 8 (88): 5143. https://doi.org/10.21105/joss.05143.

Dai, Yun. 2025. "Why Students Use or Not Use Generative AI: Student Conceptions, Concerns, and Implications for Engineering Education." *Digital Engineering* 4 (March): 100019. https://doi.org/10.1016/j.dte.2024.100019.

Emirbayer, Mustafa, and Ann Mische. 1998. "What Is Agency?" *American Journal of Sociology* 103 (4): 962–1023. https://doi.org/10.1086/231294.

Fairclough, Norman. 2013. *Critical Discourse Analysis: The Critical Study of Language*. Routledge.

Foucault, Michel. 1980. "Truth and Power." In *Power/Knowledge: Selected Interviews and Other Writings, 1972–1977*, edited by Colin Gordon, 109–33. New York: Pantheon Books.

Guest, Olivia, Marcela Suarez, Barbara Müller, Edwin van Meerkerk, Arnoud Oude Groote Beverborg, Ronald de Haan, Andrea Reyes Elizondo, et al. 2025. "Against the Uncritical Adoption of 'AI' Technologies in Academia." Zenodo. https://doi.org/10.5281/ZENODO.17065099.

Heyndels, Sybren. 2023. "Technology and Neutrality." *Philosophy & Technology* 36 (4): 75. https://doi.org/10.1007/s13347-023-00672-1.

Johnson, Simon, and Daron Acemoglu. 2023. *Power and Progress: Our Thousand-Year Struggle Over Technology and Prosperity*. Hachette UK.

Kramer, Martin, and Ritva Engeström. 2019. "Teachers' Beliefs as a Component of Motivational Force of Professional Agency." *Learning, Culture and Social Interaction* 21 (June): 214–22. https://doi.org/10.1016/j.lcsi.2019.03.007.

Kumar, Harsh, Ilya Musabirov, Mohi Reza, Jiakai Shi, Xinyuan Wang, Joseph Jay Williams, Anastasia Kuzminykh, and Michael Liut. 2024. "Guiding Students in Using LLMs in Supported Learning Environments: Effects on Interaction Dynamics, Learner Performance, Confidence, and Trust." *Proc. ACM Hum.-Comput. Interact.* 8 (CSCW2): 499:1–30.

https://doi.org/10.1145/3687038.

Latour, Bruno. 1994. "On Technical Mediation. Philosophy, Sociology, Genealogy." *Common Knowledge* 3: 29–64.

Legaspi, Roberto, Zhengqi He, and Taro Toyoizumi. 2019. "Synthetic Agency: Sense of Agency in Artificial Intelligence." *Current Opinion in Behavioral Sciences*, Artificial Intelligence, 29 (October): 84–90. https://doi.org/10.1016/j.cobeha.2019.04.004.

Leonardi, Paul M. 2025. "*Homo Agenticus* in the Age of Agentic AI: Agency Loops, Power Displacement, and the Circulation of Responsibility." *Information and Organization* 35 (3): 100582. https://doi.org/10.1016/j.infoandorg.2025.100582.

Leontiev, Dmitry A. 2007. "Approaching Worldview Structure with Ultimate Meanings Technique." *Journal of Humanistic Psychology* 47 (2): 243–66. https://doi.org/10.1177/0022167806293009.

McCormick, Miriam Schleifer. 2014. *Believing Against the Evidence*. 0th ed. Routledge. https://doi.org/10.4324/9780203579146.

Morrow, David R. 2014. "When Technologies Makes Good People Do Bad Things: Another Argument Against the Value-Neutrality of Technologies." *Science and Engineering Ethics* 20 (2): 329–43. https://doi.org/10.1007/s11948-013-9464-1.

Mouta, Ana, Ana María Pinto-Llorente, and Eva María Torrecilla-Sánchez. 2025. "'Where Is Agency Moving to?': Exploring the Interplay Between AI Technologies in Education and Human Agency." *Digital Society* 4 (2): 49. https://doi.org/10.1007/s44206-025-00203-9.

Naeem, Muhammad, Wilson Ozuem, Kerry Howell, and Silvia Ranfagni. 2023. "A Step-by-Step Process of Thematic Analysis to Develop a Conceptual Model in Qualitative Research." *International Journal of Qualitative Methods* 22 (October): 16094069231205789. https://doi.org/10.1177/16094069231205789.

Naik, Ishita, Dishita Naik, and Nitin Naik. 2024. "ChatGPT Is All You Need: Untangling Its Underlying AI Models, Architecture, Training Procedure, Capabilities, Limitations And Applications." Preprints. https://doi.org/10.36227/techrxiv.173273427.76836200/v1.

Nespor, Jan. 1987. "The Role of Beliefs in the Practice of Teaching." *Journal of Curriculum Studies* 19 (4): 317–28. https://doi.org/10.1080/0022027870190403.

Pew Research Center, Janna Anderson, and Lee Rainie. 2023. "The Future of Human Agency," February.

Sartre, Jean Paul. 1946. "Existentialism Is a Humanism." https://www.marxists.org/reference/archive/sartre/works/exist/sartre.htm.

Silvennoinen, Minna, Satu Aksovaara, and Merja Alanko-Turunen. 2025. "Students' Usage of GenAI in Universities of Applied Sciences: Experiences and Development Needs for Guidance and Support." In *38th Bled eConference: Empowering Transformation: Shaping Digital Futures for All: Conference Proceedings*, 467–82. University of Maribor Press. https://doi.org/10.18690/um.fov.4.2025.29.

Skinner, Ellen A. n.d. "A Guide to Constructs of Control."

Sundar, S Shyam. 2020. "Rise of Machine Agency: A Framework for Studying the Psychology of Human–AI Interaction (HAII)." *Journal of Computer-Mediated Communication* 25 (1): 74–88. https://doi.org/10.1093/jcmc/zmz026.

Suonpää, Maija, Jutta Heikkilä, and Ana Dimkar. 2024. "Students' Perceptions Of Generative

Ai Usage And Risks In A Finnish Higher Education Institution." In *18th International Technology, Education and Development Conference*, 3071–77. Valencia, Spain. https://doi.org/10.21125/inted.2024.0825.

Suriano, Rossella, Alessio Plebe, Alessandro Acciai, and Rosa Angela Fabio. 2025. "Student Interaction with ChatGPT Can Promote Complex Critical Thinking Skills." *Learning and Instruction* 95 (February): 102011. https://doi.org/10.1016/j.learninstruc.2024.102011.

Swanepoel, Danielle. 2021. "Does Artificial Intelligence Have Agency?" In *The Mind-Technology Problem*, edited by Robert W. Clowes, Klaus Gärtner, and Inês Hipólito, 18:83–104. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-030-72644-7_4.

Winner, Langdon. 2018. "The Cult of Innovation: Its Myths and Rituals." In *Engineering a Better Future: Interplay Between Engineering, Social Sciences, and Innovation*, edited by Eswaran Subrahmanian, Toluwalogo Odumosu, and Jeffrey Y. Tsao, 61–73. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-319-91134-2_8.